1. Learning (10/1/26-11/1/26)

1.1 Study ROS2, and ROS2 system design

This is going to be the single most important thing we can do early on, it's important everyone has a grasp of the software we are using and how do design a system in it

1.2 Dev/Test Environment

Set up a development and testing environment and ensure a consistent coding style among the team

1.3 Simulations

Have the team run turtlesim and implement their own ROS2 systems to get a grasp of the tool and get everyone familiar with simulations, which will be incredibly important until we have a physical robot

2. Design (11/1/26-11/15/26)

2.1 Network Arhitecture

This will be a network diagram defining all the systems on the network, and the software they run, it should include information like operating system, IP, MAC if possible, and which other devices it connects to and with what protocols

2.2 Sequence Diagram

This will be a sequence diagram showing the proposed sequence of actions by different actors in the system and what the proposed responses would be

2.3 Module Diagram

This shows the encapsulation of the program, what state it owns and defines its API

3. Implementation (11/15/26-2/1/27)

3.1 Drive Train

3.2 Scooper

3.3 UI

4. Testing (2/1/27-)

This is incredibly important as it allows us to gather feedback from non-technical members and make adjustments to the system where needed, this section is REQUIRED, there will be bugs in the software, it will not work our first run.

5. Automation (2/1/27-)

This is single handedly the hardest part of the process, this is by no means a solved problem and require intense discussion, planning, prep, and knowledge. It will be something we are tweaking up to the day of the competition.